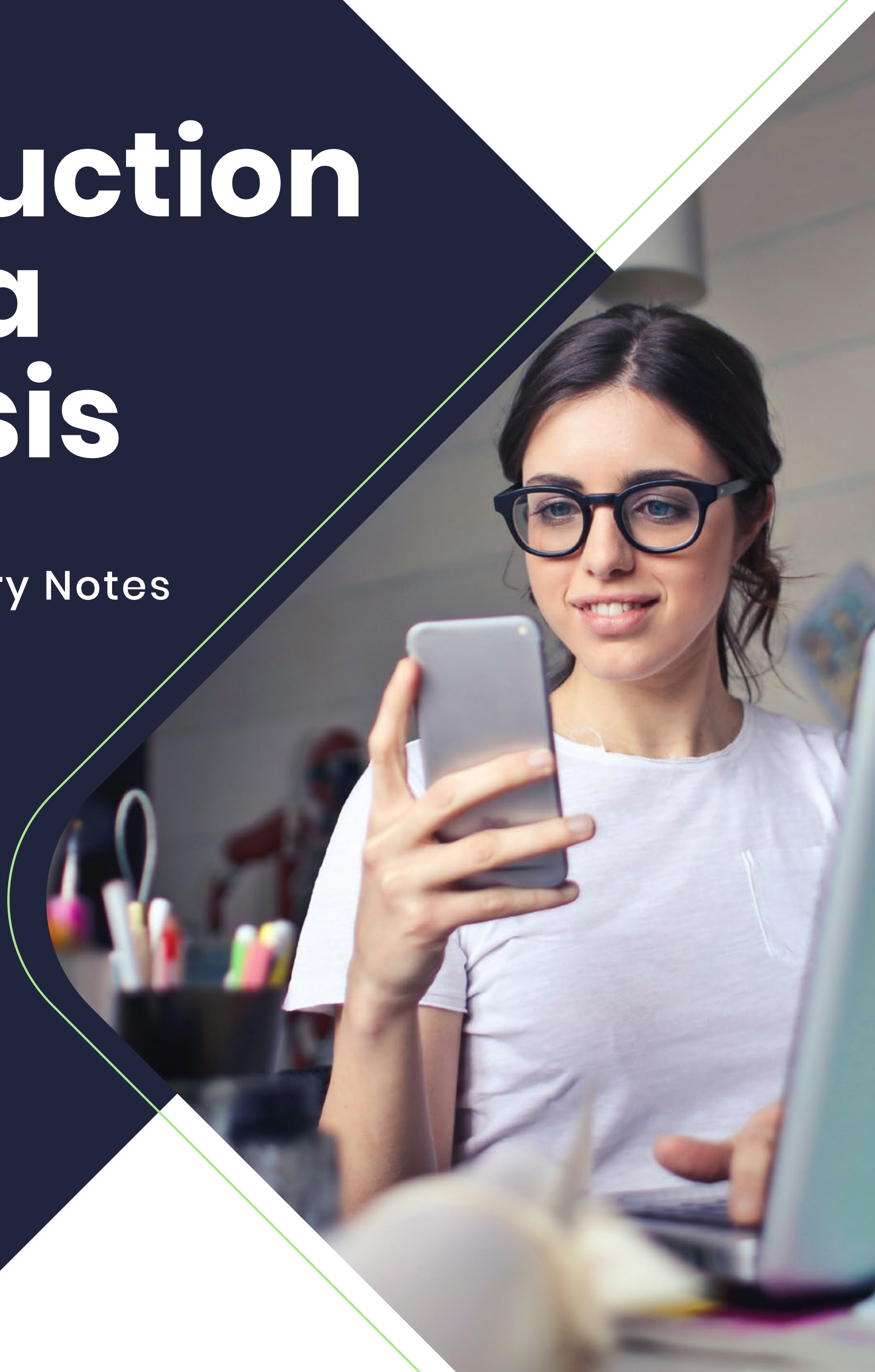
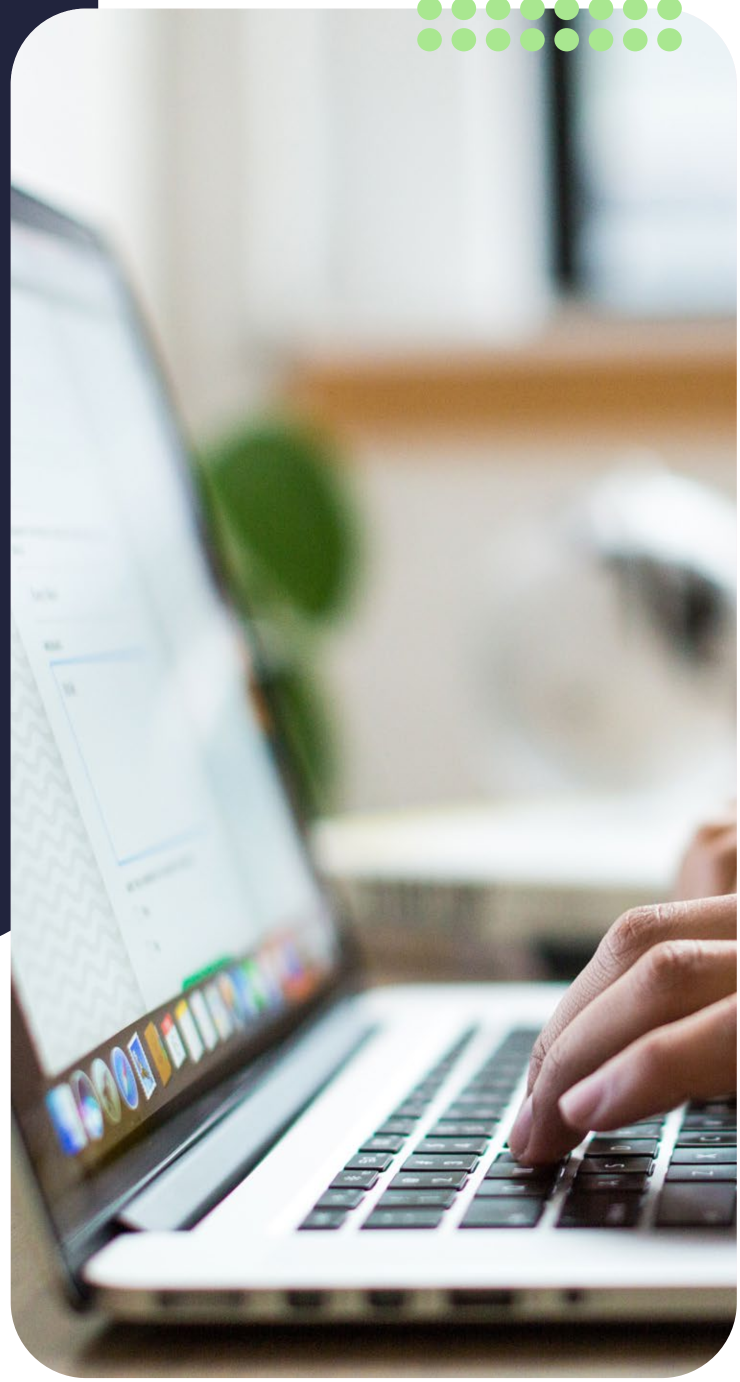
Diploma in Data Analysis

Introduction to Data Analysis

Lesson 7: Summary Notes



Contents



Lesson Objectives

- Hypothesis testing: more large sample cases
- P-value
- Hypothesis testing: small sample

Lesson Introduction

What happens when we don't have enough patients to take part in the study because we are dealing with a rare disease or if the clinical trial is very expensive and the benefactors are not able to contribute financially towards a larger sample and we cannot gather a sample larger than 30 observations or data points? Can we still use the same hypothesis test as for a large sample?

In Lesson 6, we introduced hypothesis testing for the case of a difference between 2 means with a large sample, meaning that the distribution was normally distributed.

In Lesson 7, we will continue to explore hypothesis testing for large samples for the cases where the alternative hypothesis is less or greater than the population parameter. Thereafter we will shed light on the well-known term, the p-value, that is frequently used, but not necessarily correctly. For the last topic, we will look into what happens when the sample contains less than 30 observations.

Did you know?

The concept of hypothesis testing grew out of the need for big batches of manufactured items to be accepted based on smaller samples in the quality control industry.

Hypothesis testing

Hypothesis testing steps recap

1. The first step in our test is to establish the null and alternative hypothesis.
2. Thereafter, we define the critical value that defines our rejection and acceptance rejoin.
3. Thirdly, we calculate our test statistic.
4. And finally, we draw a conclusion by either accepting or rejecting the null hypothesis.

One-tail vs two-tail

Previously, we have only explored the case of the means not being equal. This case is classified as a two-tailed test because the claim of the mean not being equal to a certain value results in the z-score on both sides of the statistic. If our hypothesis states that two groups differ, but does not indicate directions, that is, which is greater, we are dealing with a two-tailed test and the rejection region is located in both tails of the sampling distribution.
In contrast, if the hypothesis indicates directions, that is that one sample group is greater than the other for instance, then we are dealing with a one-tailed test and the rejection region is located in either the lower or upper tail of the sampling distribution.

Example

If a certain brand of chocolate bar you buy in the shop's mean weight is less than the 100g it says it weighs on the wrapper, it is an example of a one-tailed sampling distribution.

The null hypothesis would be that the mean weight is greater or equal to the value that is currently being accepted as true, 100g, and the alternative hypothesis is the claim that we want to test, that the mean weight is less than 100g.

H_o:mean ≥ 100 g
H_a:mean < 100 g

These two statements, as always, are mathematical opposites of each other.

Critical value

Remember, a large sample size means the Central Limit Theorem is valid and we use the z-test statistic as a test statistic. The normal distribution's shape always stays the same therefore, the critical values will always remain the same at the various confidence levels.

A two-tailed test allocates half of the alpha (α) value to testing the statistical significance in upper tail of the normal distribution and half of the alpha value to testing statistical significance in the lower tail of the normal distribution. If using a significance level of 0.05, then 0.025 is in each tail of the distribution of the test statistic and the critical value is +/- 1.96.

A one-tailed test, on the other hand, allocates the entire alpha to testing statistical significance in either the upper or lower tail of the distribution.

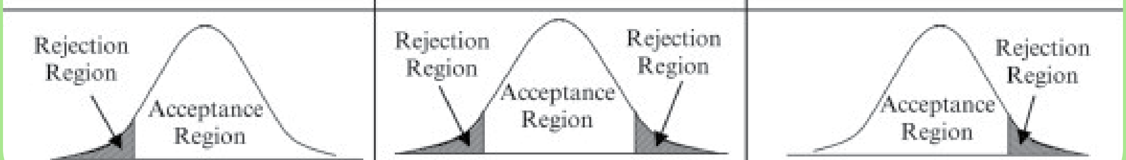
Therefore, with a significance level of 0.05, then 0.05 will be in either the upper or lower tail of the distribution. And the test statistic will be +/- 1.645 depending on the test.

Regions

For the two-tailed hypothesis, the decision was made to reject the null hypothesis if the test statistic fell in the rejection regions that are located in the upper and lower tails of the sample distribution.

For a one-tailed hypothesis test, the rejection region is located in either the upper or lower tail of the sampling distribution depending on if we want to determine if the population mean is greater or less than a given quantity.

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X < \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X \neq \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X > \mu_0$



Therefore, if the alternative hypothesis states that the population mean is less than a given quantity, and the test statistic lies in the lower tail rejection region, the null hypothesis is rejected.

Example

At a 95% confidence level (α= 0.05) for the example, we will have a critical value of -1.645 for a lower-tailed one sample hypothesis test. The rejection region, therefore, lies in the lower tail of the sampling distribution.

NOTES

Test statistic

A test statistic measures the degree of agreement between a sample of data and the null hypothesis.
For the normal distribution, we use the variable z as the test statistic for where the population standard deviation is known.

The formula for the z test statistic is:

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where

$\bar{X}$ = sample mean
μ= population mean
S = population standard deviation
$\sqrt{n}$ = sample size
If the test statistic lies in the rejection region, meaning if it's passed the critical value, we would reject the null hypothesis and say that the null and alternative hypothesis is more likely not the same.

Example

If we sample 30 bars with a mean weight of 99g from the chocolate bars sample, a population standard deviation of 1 and a population mean of 100g, inputting these values into the formula would result in a test statistic of -5.478.
$\bar{x}$=99, μ = 100, σ = 1, n = 30

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$z = \frac{99 - 100}{0{,}1/\sqrt{30}}$$

z = -5.478

Conclusion

The last step would be to either reject the null hypothesis and accept the alternative hypothesis or to not reject the null hypothesis.

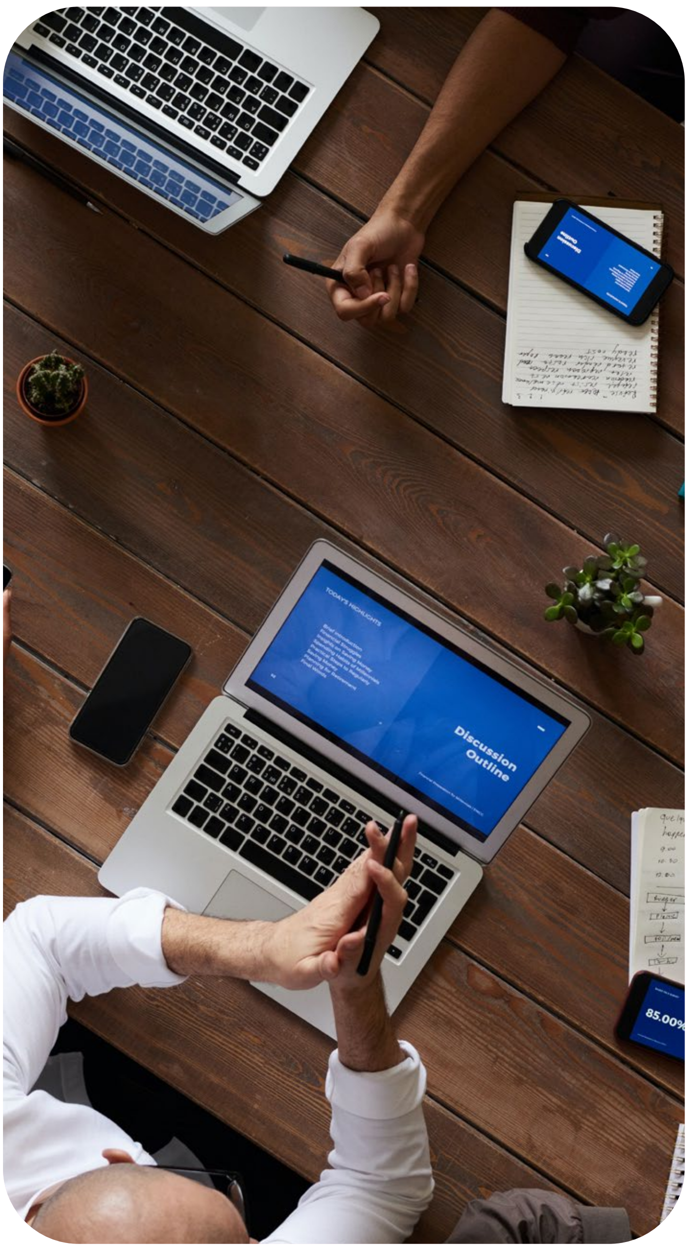
Do we reject or not reject the null hypothesis for this example?

Errors

Are the results of the hypothesis tests reliable?

We as the researcher have to decide on the significance level before we analyse the data, the choice of the significance level is based on the consequences of making a type I and type I error.

To use our medical example again, if drug 1 is much more expensive than drug 2, with our null hypothesis being that the drugs are equally as effective and the alternative that drug 2 is more effective, making a type 1 error by assuming drug 2 is more effective, would result in the patient paying a lot of extra money for the drug when it is not needed. Therefore, we would want a small significance error to prevent this.



NOTES

P-value

Did you know?

In March 2016, the American Statistical Association released a statement to warn about the use and interpretation of p-value.

P-value defined

The p-value or probability value indicates Statistical significance in a hypothesis test.
The p-value assumes that the null hypothesis is true and tells us what the probability is of observing data as extreme as the actual observed data. So it tells us how incompatible the data is with a chosen statistical model. Admond Lee said that the lower the p-value, the more surprising the evidence is, the more ridiculous our null hypothesis looks. We feel ridiculous with our null hypothesis because we rejected that and choose our alternative hypothesis instead [sic].

The p-value is not the probability that the data were produced by random chance alone.

Relationship between the p-value and Type 1 error

The Type 1 error is when the null hypothesis is rejected when it should not have been, also known as a false positive. Whereas a p-value is the probability of the observed data given that the null hypothesis is true.

Interpreting the p-value

Let the p-value cut-off value be 0.05, a convention widely accepted for the p-value, we interpret a large p-value greater than 0.05 ($p > 0.05$) to be weak evidence against the null hypothesis, so you fail to reject the null hypothesis. On the other hand, if the p-value was equal or smaller than 0.05 ($p < 0.05$), then we would interpret the p-value to indicate strong evidence against the null hypothesis, so you reject the null hypothesis.

A p-value close to the cut off line so to speak is interpreted as marginally significant.

Statistical significance

If the earlier example of the 2 drugs hypothesis test generated a p-value of 0.01, and we found that the null hypothesis to be true, that the drugs do not have a different effect on its ability to cure the patient, we would say that based on the p-value, when looking at the population as a whole, 1% of studies will procure the effect observed in your sample because of random sample error.

Remember

We have previously learned that descriptive statistics is the terms that we use to describe and summarise the data whereas inferential statistics allows us to make conclusions about any, for example, hypothesis that we have made. When studying data, it is important to describe the data with descriptive statistics but also interpret other indexes, like the p-value, to as a whole correctly study the data.

NOTES

Hypothesis testing for smaller samples

Small sample hypothesis test defined

When we have a small sample with less than 30 observations, the Central Limit Theorem does not apply and we have to apply stricter assumptions in the form of a t-test to ensure statistical validity to the testing procedure.

Test statistic

Let's investigate the case where the for a small sample hypothesis test with a single population mean where the population standard deviation is unknown with a sample standard deviation of s. Our test statistic would follow the student's t-distribution with degrees of freedom (d.f.) equal to the number of observations in the sample, called n-1

Example

Let's say we have a sample of 5 chocolate with weights 99 g, 98 g, 99.5 g, 99.6 g, 100 g
The mean weight of the 5 chocolate bars is 99.22 g.

$$\mu = \frac{\sum X}{n}$$

Step 1

If the null hypothesis is that the mean weight for the population is 99.2g grams, the same as that of our sample, we can choose the alternative hypothesis that it is not equal to the mean weight, so not equal to 99.2g.

$$H_0: \mu = 99.2\ g$$

$$H_a: \mu \neq 99.2\ g$$

Step 2

To test this statement, let's select a total of 20 chocolate bars from the shop at an alpha value of 0.05.
We calculate a critical t-value of 2.093 from our sample data.

Next, we calculate the regions that we will compare our test statistic t to, in order to know if we need to reject the null hypothesis or not.

We have a two-tailed test, therefore the rejection region will be alpha divided by two (α/2) for the half of alpha that is in the upper tail and negative alpha divided by two for the lower tail.

The decision rule would be:
If t is less than -3.11 or greater than 3.11, we would reject the null hypothesis

Step 3

Let's assume that the population mean is 100g as indicated on the chocolate wrapper and the standard deviation is 1g.

To calculate the test statistic, we use the formula we have previously seen to plug the values of the population mean (x = 99.2 g), sample mean (μ = 100 g), sample standard deviation (s = 1) and sample observations (n = 20) into the equation.

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$z = \frac{99{,}2 - 100}{1/\sqrt{20}}$$

t = -3.578

Step 4

The next step would be to state the results.

We reject the null hypothesis if t is less than -3.11 or greater than 3.11. Our t-value is less than -3.11, therefore we reject the null hypothesis.

There is a significant difference between the weight of our sample and the assumed population mean of 100 g.

NOTES

Resources

Beers, B., 2020, P-Value Definition, Investopedia, https://www.investopedia.com/terms/p/p-value.asp

Bevans, R., 2020, An introduction to t-tests, Scribbr, https://www.scribbr.com/statistics/t-test/

Frost, J., 2018, Interpreting P values, Statistics By Jim, https://statisticsbyjim.com/hypothesis-testing/interpreting-p-values/

Frost, J., 2018, One-Tailed and Two-Tailed Hypothesis Tests Explained, Statistics By Jim, https://statisticsbyjim.com/hypothesis-testing/one-tailed-two-tailed-hypothesis-tests/

Kitchin, J., 1994, 'Rejection Region', Methods in Experimental Physics, https://www.sciencedirect.com/topics/mathematics/rejection-region

LaMorte, W.W., 2019, One Sample t-test, Boston University School of Public Health, https://sphweb.bumc.bu.edu/otlt/MPH-Modules/PH717-QuantCore/PH717-Module7-T-tests/PH717-Module7-T-tests4.html

Lee, A., 2019, P-values Explained By Data Scientist, towards data science, https://towardsdatascience.com/p-values-explained-by-data-scientist-f40a746cfc8

Lund, A., Lund, M., Descriptive and Inferential Statistics, Laerd statistics, https://statistics.laerd.com/statistical-guides/descriptive-inferential-statistics.php

Privitera, G.J. (2011). Essential Statistics for the Behavioural Sciences [online]. SAGE Publications, Inc. https://www.sagepub.com/sites/default/files/upm-binaries/40007_Chapter8.pdf

Rumsey, D.J., What a p-value Tells You about Statistical Data, dummies, https://www.dummies.com/education/math/statistics/what-a-p-value-tells-you-about-statistical-data/#:~:text=The%20p%2Dvalue%20is%20a,you%20reject%20the%20null%20hypothesis.&text=p%2Dvalues%20very%20close%20to,(could%20go%20either%20way).

Smith, M.K., 2011, Type I and II Errors and Significance Levels, University of Texas, https://web.ma.utexas.edu/users/mks/statmistakes/errortypes.html

Tanha, K., Mohammadi, N., Janani, L., 2017, P-value: What is and what is not, Medical Journal of

The Islamic Republic of Iran, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5804470/
University of California, 2020, What are the differences between one-tailed and two-tailed tests?, https://stats.idre.ucla.edu/other/mult-pkg/faq/general/faq-what-are-the-differences-between-one-tailed-and-two-tailed-tests/